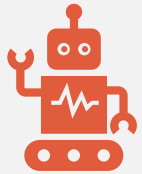


What is Mechatronics?

9 Computing Technology

Learning Intention

Describe mechatronic systems



Investigate the development and impact of a mechatronic, robotic
and/or automated system in a real-world application

Success Criteria



I can describe mechatronic components



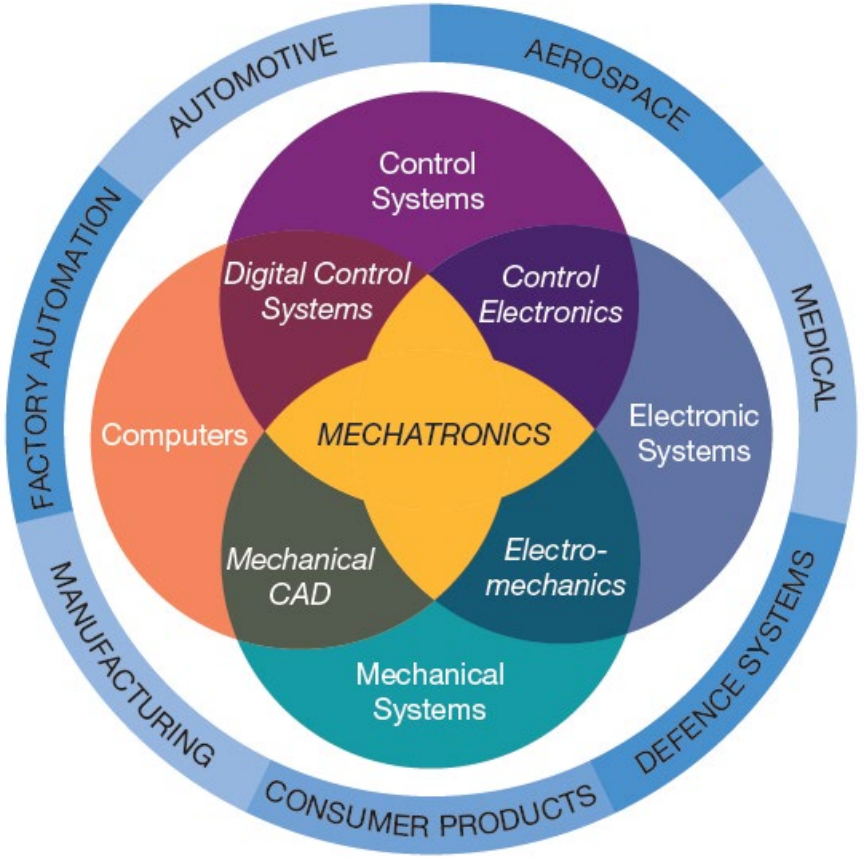
I can provide examples of mechatronic systems



I can distinguish between mechatronics and robotics

Mechatronics, or mechatronic engineering, is a multidisciplinary field combining work done in disciplines such as mechanical, electrical, electronics, computer science, and software engineering.

Introduction to Mechatronics



- Mechatronic devices = “smart machines”
- Combine mechanical, electronic & software engineering
- Examples: robots, drones, washing machines
- Engineers design devices that respond to their environment

Mechatronics in Daily Life



Escalators speed up when stepped on



Smart lighting in carparks & stairwells



Doorbells with video/audio notifications



Dishwashers & air-conditioners monitor usage



Cars track maintenance automatically

History & Terminology

Term ‘mechatronics’ coined in 1969 by Tetsuro Mori (Yaskawa)

Combination of **mechanics + electronics**

Basis for IoT, STEM projects, and the maker movement

Mechatronics in Industry

Control assembly lines, factory automation

Functions include:

Welding & spray-painting	Assembly operations	Palletising & material handling	Dispensing operations	3D printing & lab automation

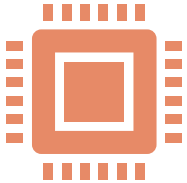
Key Components of Mechatronic Systems



Sensors: measure environment & send data



Actuators: perform mechanical actions (effectors)



Microcontrollers: process data & control actuators



Often include **feedback loops** for adjustments

Microcontrollers



Small, dedicated computers on a single board



Interact with: accelerometers, motors, pumps, sensors



Cheap, low memory, slow but easy to interface with real-world devices

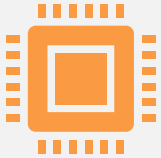


Examples: **micro:bit**, Arduino

Applications in Homes & Schools

- Printing school tasks
- Flying drones
- Smoothie blenders & coffee machines
- Apple Watch handwashing detection (motion + audio)
- Reverse vending machines (Recycling)

Mechatronics & System Design



Systems thinking: combine mechanical, electronic & computing technologies



Design considers: efficiency, sustainability, and automation



Security, reliability, and user interaction often included

Is Mechatronics another word for Robotics?

No!

Robotics is an interdisciplinary branch of engineering and computer science that involves the design, construction, operation, and application of robots



Mechatronics, or mechatronic engineering, is a multidisciplinary field combining work done in disciplines such as mechanical, electrical, electronics, computer science, and software engineering.

